DSA Mastery Roadmap

Target: Solve 200–250 high-quality problems to achieve complete DSA mastery.

0. Mathematical & Analytical Foundations

- Asymptotic Analysis (Big-O, Big-Ω, Big-Θ)
- Time vs Space Tradeoffs
- Recurrence Relations & Master Theorem
- Proof Techniques (Induction, Contradiction)
- Basic Probability (Randomized Algorithms)

1. Basic Data Structures

- Arrays (Prefix Sum, Sliding Window, Kadane)
- Strings (KMP, Rabin-Karp, Z-Algorithm)
- Linked Lists (Cycle Detection, Reversal, Merging)

2. Core Abstract Data Types

- Stack (Monotonic Stack, Expression Evaluation)
- Queue & Deque
- Hashing (Collision Handling, Rolling Hash)

3. Trees

- Binary Trees (DFS, BFS, LCA, Diameter)
- Binary Search Trees
- Balanced Trees (AVL, Red-Black, B-Trees)
- Heap (Priority Queue, Heap Sort, Top-K)
- Advanced Trees (Segment Tree, Fenwick Tree, Trie, Suffix Structures)

4. Graphs

- Graph Representations
- BFS & DFS
- Shortest Path (Dijkstra, Bellman-Ford, Floyd-Warshall)
- Minimum Spanning Tree (Kruskal, Prim, DSU)
- SCC, Topological Sort, Cycle Detection
- Network Flow & Matching

5. Recursion & Backtracking

- Subsets & Permutations
- N-Queens
- Sudoku Solver
- Combinatorics Generation

6. Dynamic Programming

- Knapsack (0/1 & Unbounded)
- Coin Change
- LIS, LCS
- DP on Trees & Graphs
- Bitmask DP
- Interval & Digit DP

7. Searching & Sorting

- Merge Sort, Quick Sort, Heap Sort
- Counting & Radix Sort
- Binary Search & Variations

8. Greedy Algorithms

- Activity Selection
- Huffman Coding
- Interval Scheduling
- Fractional Knapsack

9. Divide & Conquer

- Merge Sort
- Quick Sort
- Closest Pair of Points
- Strassen’s Matrix Multiplication

10. Randomized Algorithms

- Randomized QuickSort
- Reservoir Sampling
- Monte Carlo vs Las Vegas

11. Advanced Competitive Topics

- Heavy Light Decomposition
- Mo’s Algorithm
- Persistent Segment Trees
- Treaps
- Convex Hull Trick
- Computational Geometry
- Aho-Corasick

12. Algorithm Design Paradigms

- Greedy
- Dynamic Programming
- Divide & Conquer
- Backtracking

- Branch & Bound
- Amortized Analysis

13. Theoretical Computer Science

- NP Completeness
- P vs NP
- Reductions
- Approximation Algorithms
- Online Algorithms